

ST LAWRENCE, Australia

WMO: 953690

Lat: 22.3472S

Lon: 149.5242E

Elev: 32

StdP: 14.68

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	46.0	48.7	27.6	21.6	63.7	31.5	25.9	61.9	16.4	67.5	14.9	67.2	2.2	180	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	91.8	75.8	90.2	75.4	88.8	74.9	80.1	86.9	79.0	85.7	78.1	84.6	9.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.3	147.0	83.6	77.2	141.8	82.9	76.2	137.0	82.2	43.8	87.1	42.7	85.9	41.6	84.5	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	15.1	13.8	40.7	97.7	2.7	3.2	38.7	100.0	37.1	101.9	35.6	103.7	33.6	106.1		
(5)			37.4	82.4	2.6	1.8	35.5	83.7	34.0	84.8	32.5	85.8	30.6	87.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.2	80.9	80.6	78.9	74.8	69.5	65.0	63.7	65.0	69.6	74.0	77.3	80.1		
	DBStd	7.07	2.65	2.80	2.61	2.93	3.83	4.31	4.00	3.73	3.49	3.41	2.90	2.95		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	184	0	0	0	0	11	52	72	45	4	0	0	0		
	CDD50	8483	958	856	897	745	604	452	424	464	587	745	819	932		
	CDD65	3192	493	436	432	296	149	53	31	44	141	281	369	467		
	CDH74	27203	5097	4328	3815	1967	752	192	132	259	903	1979	3097	4682		
	CDH80	8387	1888	1577	1262	469	73	6	4	7	109	401	904	1687		
(14) Wind	WSAvg	5.9	6.7	6.1	5.9	5.3	4.8	4.8	4.8	5.2	6.0	7.1	7.3	7.1		
(15) Precipitation	PrecAvg	39.0	6.7	6.7	5.1	2.1	2.3	1.2	1.0	1.1	0.9	1.7	3.6	5.8		
	PrecMax	68.2	25.3	26.7	20.2	8.7	11.8	10.4	4.8	8.6	5.5	5.2	16.9	24.1		
	PrecMin	15.0	0.0	0.5	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	14.1	5.9	6.6	5.3	2.2	2.4	1.6	1.2	1.7	1.3	1.6	3.3	5.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	93.6	92.3	88.4	84.2	80.5	79.6	80.7	86.8	89.0	92.2	94.2		
		MCWB	78.0	76.8	75.3	72.1	68.1	66.8	63.5	63.5	65.8	70.0	69.3	75.8		
	2%	DB	90.9	91.2	89.8	86.3	82.0	78.2	77.0	78.3	82.7	85.4	87.9	90.5		
		MCWB	77.2	76.0	74.5	71.0	67.6	65.4	63.2	63.7	66.0	71.4	72.3	75.9		
	5%	DB	89.4	89.5	88.0	84.6	80.4	76.3	75.4	76.9	80.7	83.7	86.3	88.8		
		MCWB	76.2	75.8	74.2	70.5	66.7	64.6	62.5	63.5	66.7	70.5	72.1	75.0		
	10%	DB	87.8	87.8	86.4	82.9	78.5	74.4	73.6	75.2	79.0	82.2	84.9	87.3		
		MCWB	75.2	75.5	73.7	70.1	66.1	63.4	61.5	62.3	66.0	69.5	71.8	74.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.2	81.1	80.6	77.6	74.0	71.5	69.9	70.3	73.5	76.4	78.3	81.0		
		MCDB	89.2	87.5	86.9	83.2	79.1	75.2	74.4	75.6	80.2	83.3	85.8	87.4		
	2%	WB	80.0	79.7	78.7	75.5	71.6	69.3	67.8	68.6	71.6	74.8	76.4	79.4		
		MCDB	87.3	86.3	84.3	81.2	77.6	74.1	72.7	74.2	77.8	81.1	83.3	86.3		
	5%	WB	78.8	78.7	77.3	73.8	69.9	67.5	66.0	67.1	70.2	73.6	75.1	78.0		
		MCDB	85.5	85.1	83.4	80.1	76.4	73.1	71.4	72.8	76.6	80.1	82.4	85.0		
	10%	WB	77.7	77.7	76.1	72.4	68.5	66.0	64.2	65.4	68.9	72.2	73.9	76.8		
		MCDB	84.3	84.1	82.5	79.1	75.1	72.0	70.3	71.7	75.4	79.0	81.3	83.7		
(35) Mean Daily Temperature Range	MDBR		14.0	14.0	15.1	16.8	18.7	18.9	20.9	21.5	20.0	17.1	15.4	14.5		
	5% DB	MCDBR	16.1	16.8	17.4	18.8	19.8	20.0	21.8	22.5	21.3	17.9	16.9	16.0		
		MCWBR	5.3	5.3	5.6	6.7	8.2	9.6	10.3	10.5	9.3	7.0	6.1	5.4		
	5% WB	MCDBR	13.7	13.6	13.6	14.6	16.1	15.8	16.2	17.0	16.4	14.8	14.2	13.8		
		MCWBR	5.8	5.6	5.8	6.3	7.4	8.8	8.8	9.0	7.8	6.7	6.0	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.404	0.392	0.373	0.354	0.335	0.323	0.313	0.324	0.357	0.386	0.405	0.403		
	taud		2.482	2.519	2.560	2.591	2.588	2.620	2.626	2.581	2.486	2.435	2.414	2.457		
	Ebn at Noon		298	298	295	288	282	279	287	294	296	296	296	299		
	Edh at Noon		37	35	33	29	27	25	26	29	35	38	40	38		
(44) All-Sky Solar Radiation	RadAvg		2029	1921	1761	1546	1332	1168	1317	1596	1928	2136	2188	2177		
	RadStd		146	200	151	155	77	104	98	107	140	130	251	224		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	-0.85	-0.76	-0.81	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page